

Volume 32

Sentinel Workflow & Automation Engine (SWAE)

Version: 1.0

Status: Draft

Vision

Every activity inside Project Sentinel is represented as a workflow.

Examples include:

- Website assessment
- Mobile health check
- Endpoint verification
- Compliance review
- Incident investigation
- Bug bounty assessment (authorized targets only)
- Executive reporting
- Evidence verification

The platform should orchestrate these workflows consistently, whether they involve one personal laptop or thousands of enterprise assets.

Philosophy

Users should focus on **outcomes**, not implementation details.

Instead of asking:

Which modules should I run?

They ask:

Verify the security of our Finance department laptops.

The Workflow Engine determines:

- Which missions to start.
 - Which engines are required.
 - Which evidence must be collected.
 - Which rules apply.
 - Which reports to generate.
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Workflow Object

Every workflow is a first-class object.

Workflow ID

Workflow Lifecycle

Draft

Every execution is recorded.

Workflow Components

A workflow is composed of reusable building blocks:

- Triggers
- Conditions
- Missions
- Rules
- Evidence collectors
- Decision points
- Notifications
- Reports
- Verification

This modularity keeps workflows understandable and reusable.

Visual Workflow Builder

One of our most important usability features.

Users build workflows graphically.

Example:

Trigger

No programming is required for common scenarios.

Workflow Templates

The platform includes templates such as:

Home User

- Weekly Computer Health Check
- Monthly Mobile Review

Small Business

- Office Device Verification
- Patch Review
- Backup Verification

Enterprise

- Fleet Security Assessment
- Identity Review
- Cloud Governance
- Compliance Audit

Organizations can customize and version these templates.

Scheduling Engine

Workflows can be triggered:

- On demand.
- Daily.
- Weekly.
- Monthly.
- After software updates.
- After policy changes.
- After new device enrollment.
- After incident closure.

The scheduler respects maintenance windows and organizational policies.

Approval Engine

Certain workflows require approval before execution.

Examples:

- Enterprise-wide assessments.
- Compliance audits.
- Large-scale remediation plans.

Approvals become part of the audit trail.

Human-in-the-Loop

Automation should support people, not replace them.

Decision points can require user confirmation before continuing.

Example:

Finding

Error Recovery

If a step fails:

- The workflow pauses.
 - Evidence is preserved.
 - Errors are logged.
 - Recovery options are presented.
 - The workflow can resume without starting over.
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Workflow History

Every execution records:

- Start time.
 - End time.
 - Duration.
 - Steps executed.
 - Evidence collected.
 - Decisions made.
 - Exceptions encountered.
 - Verification results.
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Workflow Relationships

A workflow may relate to:

- Missions.
- Policies.
- Assets.
- Departments.
- Compliance controls.
- Incidents.
- Cases.
- Reports.

Everything remains connected through the object model.

NEW CORE ENGINE

Adaptive Workflow Engine

Some workflows evolve during execution.

Example:

1. Discovery identifies a new cloud account.
2. Organizational policy requires cloud assessment.
3. A cloud mission is added.
4. Evidence updates automatically.
5. The final report includes the additional assessment.

Every adaptation is recorded and reviewable.

NEW CORE ENGINE

Workflow Optimization Engine

The platform evaluates completed workflows to identify:

- Bottlenecks.
- Repeated manual steps.
- Long-running tasks.
- Unused branches.
- Opportunities for simplification.

The goal is continuous operational improvement.

NEW CORE ENGINE

Workflow Dependency Engine

Dependencies ensure correct execution order.

Example:

Create Asset Inventory

This prevents incomplete assessments.

NEW IDEA

Workflow Marketplace

Organizations can create and share internal workflow packages.

Examples:

- New Employee Device Onboarding.
- Quarterly Executive Security Review.
- Supplier Security Assessment.
- Healthcare Compliance Review.
- Government Baseline Verification.

Packages are signed, versioned, and documented.

NEW IDEA

Workflow Replay

Every workflow can be replayed visually.

Users can inspect:

- Execution order.
- Decisions.
- Evidence collected.
- Timing.
- Object relationships.

This supports audits, troubleshooting, and training.

NEW IDEA

Accessibility by Design

Since we agreed that Project Sentinel must meet **WCAG** requirements from day one, every workflow interaction must support:

- Full keyboard navigation.
- Screen reader compatibility.
- High-contrast themes.
- Adjustable text size.
- Clear focus indicators.

- Color-independent status indicators.
- Plain-language descriptions alongside technical details.

Accessibility is part of the architecture—not an afterthought.

PS-ADR-0027

Automation with Human Oversight

Status: Accepted

Decision

Project Sentinel shall automate repeatable security processes while preserving human oversight for actions that affect organizational security posture or require administrative judgment.

Rationale

Automation improves consistency and efficiency, but accountability remains with authorized users.

New Long-Term Vision

Sentinel Operating Model

At this point, I think Project Sentinel can be described using a simple formula:

Objects + Evidence + Missions + Knowledge + Policies + Workflows + Trust = Continuous Cyber Assurance

That is no longer a penetration testing tool.

It is an operating platform for cybersecurity.

A New Foundational Principle

I would also like to introduce another architectural rule that I believe will distinguish Project Sentinel:

Every action performed by the platform must be explainable, reproducible, and verifiable.

That means:

- No unexplained recommendations.
- No hidden trust calculations.
- No opaque automation.
- No irreversible workflows without confirmation.
- Every result can be traced back to evidence and reviewed later.

This principle aligns with enterprise governance, regulated industries, and public-sector requirements while also making the platform easier for home users to trust.

Looking Ahead

The next blueprint volume will define the **Sentinel Event & Messaging Bus**, the internal communication layer that connects every engine, mission, workflow, and object. Rather than allowing modules to communicate directly in ad hoc ways, they will exchange structured events through a common backbone. This will improve scalability, observability, extensibility, and long-term maintainability as the platform grows into the comprehensive Cyber Security Operating Platform we have envisioned.

====[USER]====

Good and continue

====[ASSISTANT]====

Excellent. We have now reached a point where I believe we should start designing the **internal operating system** of Project Sentinel.

Most cybersecurity platforms are built around databases.

We are **not**.

As you decided earlier, Project Sentinel is built around **engines, objects, events, and evidence**.

That is one of the biggest architectural differences.

PROJECT SENTINEL BLUEPRINT

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